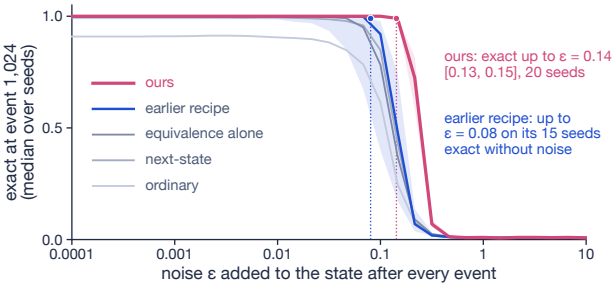
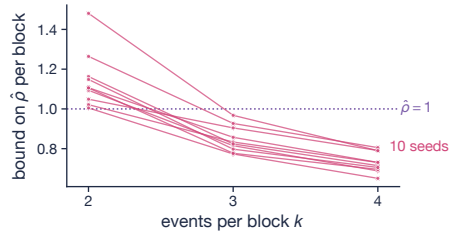
a **Ours tolerates more noise than our earlier recipe**

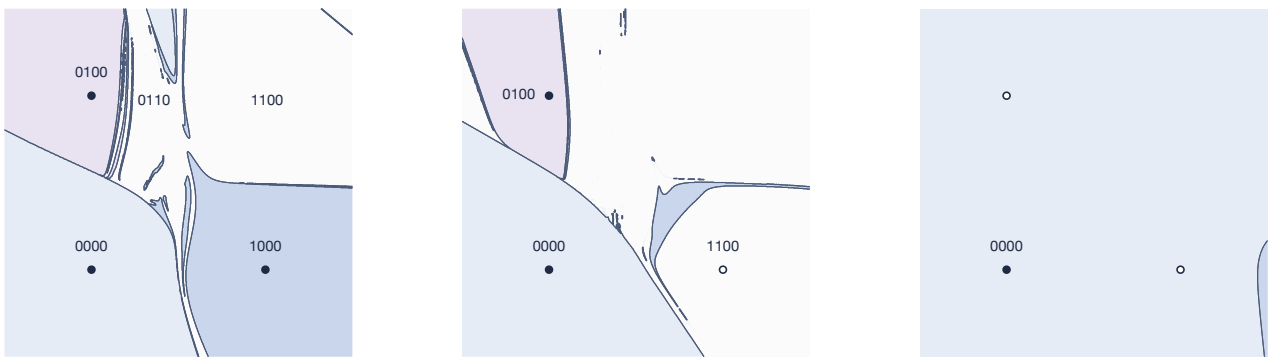


b **The automata snapped from the Lean trackers are verified exact**



snapped automata verified exact on 10 of 10 seeds; the continuous trackers' bound is below 1 from blocks of 3 events, and 0 of 38,400 repeated rotations fail; their linearised ϕ_{max} is 0.20–0.27 on 3 seeds, needs ≥ 1

c **The earlier recipe keeps every centre's basin; ordinary training loses two of three**



Code execution, seed 0; 1,001 × 1,001 start states on the plane through the centres of 0000, 1000 and 0100, each run for 64 random class-preserving cycles and coloured by the class it reports; a dot is filled when the centre's own run keeps its class

d **Lean: a perturbation dies within ten rewrites under the earlier recipe and stays under ordinary training**

